# ESP32-S3 AI Camera Module

SKU: DFR1154

The ESP32-S3 AI CAM is an intelligent camera module designed around the ESP32-S3 chip, tailored for video image processing and voice interaction. It is suitable for AI projects such as video surveillance, edge image recognition, and voice dialogue. The ESP32-S3 supports high-performance neural network computation and signal processing, equipping the device with powerful image recognition and voice interaction capabilities.

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## Specification

### Basic Parameters

| Parameter | Value |
|---|---|
| Operating Voltage | 3.3V |
| Type-C Input Voltage | 5V DC |
| VIN Input Voltage | 3.7-15V DC |

| Parameter | Value |
|---|---|
| Operating Temperature | -10~60℃ |
| Module Dimensions | 42*42mm |

## Camera Specifications

| Parameter | Value |
|---|---|
| Sensor Model | OV3660 |
| Pixel | 3 Megapixels |
| Sensitivity | Visible light, 940nm infrared |
| Field of View | 160° |
| Focal Length | 0.95 |
| Aperture | 2.0 |
| Distortion | <8% |

## Hardware Information

| Parameter | Value |
|---|---|
| Processor | Xtensa® Dual-core 32-bit LX7 Microprocessor |
| Main Frequency | 240 MHz |
| SRAM | 512KB |
| ROM | 384KB |
| Flash | 16MB |
| PSRAM | 8MB |
| RTC SRAM | 16KB |
| USB | USB 2.0 OTG Full-Speed Interface |

**Wi-Fi**

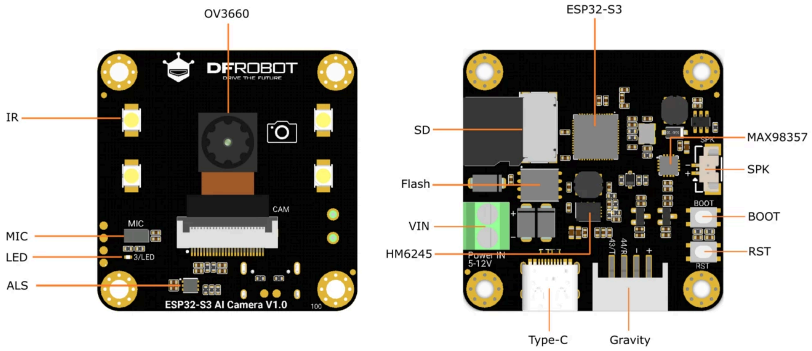
| Parameter | Value |
|---|---|
| Wi-Fi Protocol | IEEE 802.11b/g/n |
| Wi-Fi Bandwidth | 2.4 GHz band supports 20 MHz and 40 MHz bandwidth |
| Wi-Fi Modes | Station Mode, SoftAP Mode, SoftAP+Station Mode, and Promiscuous Mode |
| Wi-Fi Frequency | 2.4GHz |
| Frame Aggregation | TX/RX A-MPDU, TX/RX A-MSDU |

**Bluetooth**

| Parameter | Value |
|---|---|
| Bluetooth Protocol | Bluetooth 5, Bluetooth Mesh |
| Bluetooth Data Rates | 125 Kbps, 500 Kbps, 1 Mbps, 2 Mbps |

## Pinout

### On-board Function Diagram




### Pin & Component Description

| Label | Description |
|---|---|
| **OV3660** | 160° wide-angle infrared camera |
| **IR** | Infrared illumination (GPIO47) |
| **MIC** | I²S PDM microphone |
| **LED** | Onboard LED (GPIO3) |
| **ALS** | LTR-308 ambient light sensor |
| **ESP32-S3** | ESP32-S3R8 chip |
| **SD** | SD card slot |
| **Flash** | 16 MB Flash |
| **VIN** | 3.7–15 V DC input |
| **HM6245** | Power management chip |
| **Type-C** | USB Type-C interface for power supply and firmware upload |
| **Gravity (+)** | V1.0: 3.3–5 V input<br>V1.1: 3.3 V output |
| **Gravity (-)** | GND |
| **Gravity 44** | GPIO44 (native RX on ESP32-S3) |
| **Gravity 43** | GPIO43 (native TX on ESP32-S3) |
| **RST** | Reset button |
| **BOOT** | BOOT button (GPIO0) |
| **SPK** | MX1.25-2P speaker interface |
| **MAX98357** | I²S audio amplifier chip |

### On-board Function Pin Definition

| CAM | GPIO | MIC | GPIO | MAX98357 | GPIO | ALS | GPIO | SD | GPIO | Other | GPIO |
|---|---|---|---|---|---|---|---|---|---|---|---|
| XCLK | 5 | PDM_CLK | 38 | BCLK | 45 | SDA | 8 | MO | 11 | LED | 3 |
| Y9 | 4 | PDM_DATA | 39 | LRCLK | 46 | SCL | 9 | MI | 13 | IR | 47 |
| Y8 | 6 | | | DIN | 42 | INT | 48 | SCLK | 12 | BOOT | 0 |
| Y7 | 7 | | | GAIN | 41 | | | CS | 10 | SDA | 8 |
| Y6 | 14 | | | MODE | 40 | | | | | SCL | 9 |
| Y5 | 17 | | | | | | | | | TX | 43 |
| Y4 | 21 | | | | | | | | | RX | 44 |
| Y3 | 18 | | | | | | | | | | |
| Y2 | 16 | | | | | | | | | | |
| VSYNC | 1 | | | | | | | | | | |
| HREF | 2 | | | | | | | | | | |
| PCLK | 15 | | | | | | | | | | |
| SIOD | 8 | | | | | | | | | | |
| SIOC | 9 | | | | | | | | | | |

## FAQ

**Are there any customer reviews or in-depth evaluations available for the ESP32-S3 AI Camera Module?**

**What protocol does the MIC use?**

**What is the specification of the speaker interface? What parameters of speaker are supported?**

**Can the EdgeImpulse object classification model be customized?**

**What are the parameters of the infrared fill light?**

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